Learned Multi-Channel Fusion of Multilingual Emergency Calls and SMS Reports: Cross-Script Place Matching, Evidence Judgement and Cost-Aware Incident Merging

***Abstract*—Emergency services in Sri Lanka receive flood and fire reports by voice call and SMS in Sinhala, Tamil, English and code-mixed Singlish and Tanglish. One emergency is often reported several times, while separate emergencies can share a road, a type and a time. Merging two real emergencies can leave one without a response, whereas recording one emergency as several wastes scarce resources. This paper presents a multi-source incident fusion component that decides, for each incoming report, whether it belongs to an open incident or describes a new one. The component reconciles three extraction sources while keeping the caller's words verbatim, scores the report against every open incident on four independently learned channels (cross-script place comparison, per-type temporal decay, a fine-tuned LaBSE encoder and a fine-tuned mDeBERTa-v3 evidence judge that separates contradictions from count differences), and fuses them into a calibrated probability with gradient boosting. A report is merged only when this probability exceeds a threshold derived from a 3:1 cost ratio of false merges to false splits, and ambiguous cases are flagged for human review. All choices were made on a development split with adoption rules written in advance, and the test split was reported once. On a purpose-built corpus of 535 reports from 175 incidents, the live configuration wrongly merged 0.9% of test reports with a merge precision of 98.1% and linked 70.7% of repeat reports, or 84.0% at 1.7% false merges without a conservative no-location rule. Matching against whole incidents instead of single reports reduced false splits from 24.0% to 10.7%. Google Places placed only 53–58% of native-script locations within 1 km, against 83% for Latin script, which motivated learned text comparison of places. The component runs on a laptop CPU and is integrated live with the speech, dialogue and SMS triage components of the pipeline.**

Keywords—code-mixed Sinhala and Tamil, cost-sensitive decision making, emergency report fusion, evidence matching, incident linking, multilingual natural language processing, probability calibration

I. Introduction

During a flood or a fire, the same emergency may be reported by several people within minutes, through different channels and in different languages. In Sri Lanka, one house fire can arrive as a Sinhala voice call, a Tamil SMS and a Singlish message from a neighbour. Each report carries only part of the story: one gives the location, another the number of people trapped, another a later change in the situation. At the same time, separate emergencies can look almost identical. Two houses on the same road that catch fire on the same evening produce reports that share the road, the town, the type and the time, and differ only in the house number.

A dispatcher must therefore decide, for every new report, whether it belongs to an incident that is already open or describes a new one. The two possible errors have different consequences. A *false merge* attaches a report to an incident it does not belong to, and the second emergency may receive no response. A *false split* records a repeat report as a new incident, which duplicates effort but leaves the emergency visible. Duplicate-call features in dispatch software rely on a geocoded address, a single language and the call-taker's judgement [1]. None of these holds well here: geocoders are unreliable on Sinhala and Tamil script, callers mix languages and scripts freely, and during a disaster the call-taker's judgement becomes a bottleneck.

This paper describes the multi-source incident fusion component (C4) of a four-component multilingual emergency intake pipeline. The other components detect urgency in calls (C1), conduct the voice dialogue (C2) and triage SMS conversations (C3). C4 receives every finished report from C2 and C3 and maintains the set of incidents that the dispatcher sees. The work is organised around four research questions:

RQ1: Can learned models compare place descriptions across Sinhala, Tamil and Latin script well enough to support incident linking when geocoding is unreliable?

RQ2: Does an evidence-level judge that labels pairs of caller statements as corroborating, contradicting, neutral or differing only in a count add information beyond overall semantic similarity?

RQ3: Does comparing a new report with the accumulated evidence of an incident, rather than with single earlier reports, reduce false splits without increasing false merges?

RQ4: What trade-off between false merges, false splits and review load does a cost-derived threshold give, and how stable is it across data splits?

The individual ideas used here are established; the contribution lies in their combination and evaluation for this setting:

- A late-fusion design with four independently learned multilingual channels (geography, time, overall meaning and fine-grained evidence), each of which can be tested, replaced or switched off on its own.
- A rule-free cross-script place model, together with a measurement of commercial geocoding reliability by script.
- A four-way evidence judge over the caller's verbatim answers that separates count differences from real contradictions.
- Synthetic and counterfactual training rows that teach the fusion model hard constraints, such as different house numbers or a fire against a flood, without hand-written rules.
- Incident-level matching with a cost-derived threshold and an ambiguity flag, evaluated in an arrival-order replay in which the component builds its own incidents.
- A development-only protocol with adoption rules written before each run, multi-split reporting, and a

live service integrated with the other three components.

Section II reviews related work. Section III describes the dataset and the evaluation protocol, Section IV the architecture and Section V the experimental setup. Section VI presents the results and Section VII the implementation. Section VIII discusses the findings and limitations, and Section IX concludes.

II. Related Work and Research Gap

A. Record Linkage and Cost-Sensitive Decisions

Fellegi and Sunter [2] formalised record linkage as a decision over pairs of records compared field by field, with uncertain pairs sent for clerical review. Surveys of duplicate detection [3] and data matching [4] describe field comparison functions and blocking, which limits the number of comparisons. A probability supports a cost-based decision only if it is calibrated; Platt scaling [5] and later work on neural networks [6] address this. Elkan [7] showed that with a calibrated probability and asymmetric error costs, the optimal rule is a threshold set by the cost ratio, which is the rule used in this work.

B. Event Coreference and Event Tracking

Cross-document event coreference asks whether two mentions describe the same real-world event. ECB+ [8] added similar but different events so that lexical overlap is not enough, and neural models score mention pairs with learned span representations [9]. Topic detection and tracking [10] introduced the on-line setting, in which each story is assigned on arrival and compared with clusters of earlier stories. Crisis informatics showed that disaster posts are fragmented and repetitive [11] and surveyed methods to classify and cluster them [12]. These works study public posts, mostly in English, whereas emergency calls and SMS reports are private conversations guided by an operator's questions.

C. Multilingual Representations and Inference

Multilingual encoders such as XLM-R [13] and sentence encoders such as Sentence-BERT [14] made cross-lingual similarity practical; LaBSE [15] was trained so that translations receive similar embeddings and covers Sinhala and Tamil. Both languages remain low-resource [16], and romanised code-mixed text is harder still [17]. Natural language inference [18] and its cross-lingual benchmark XNLI [19], which does not include Sinhala or Tamil, provide the basis for contradiction detection, and DeBERTaV3 [20] is a strong multilingual encoder for it. Knowledge distillation [21] into mT5 [22] allows cheap multilingual extraction, and counterfactually augmented data [23] helps a model learn the features that actually matter.

D. Positioning of This Work

Table I compares the main lines of work on the requirements of this problem.

TABLE I. Positioning Against Related Work

Approach	Multi-lingual	Conflict detail	Cost-aware	Incident history
Record linkage [2]	no	field level	2 thresholds	no
Event coreference [9]	mostly English	partly	no	clusters
TDT [10]	some	no	no	yes
Crisis clustering [12]	some	no	no	partly
CAD duplicate calls	no	no	human	no
This work	**yes (5 forms)**	**evidence judge**	**3:1 ratio**	**incident level**

The gap is the combination: multilingual, code-mixed reports over voice and SMS, scored on several independent channels, decided with an explicit cost, and evaluated separately on the hard cases (the same place at another time, the same road with another house number) rather than on easy pairs.

III. Dataset and Evaluation Protocol

A. Corpus and Splits

No real calls were used. The study material is a purpose-built simulated corpus of emergency conversations that the author wrote by hand to reflect the language mix of Sri Lankan callers and the situations that make fusion difficult: several reports of one incident from different channels and languages, delayed follow-ups, and different incidents that share a place, a type or a day. Splits were made by incident, never by report, and the incidents of a trap group were kept in one split. Table II summarises the final dataset.

TABLE II. Dataset Properties

Property	Value
Incidents	175 (fire 85, flood 83, other 7)
Incident split	train 101 | dev 34 | test 40
Reports	535: 270 call, 265 SMS; train 308 | dev 112 | test 115
Language forms	Sinhala 187, Tamil 145, Singlish 81, English 79, Tanglish 43
Turns / evidence items	3,501 turns; 1,445 caller answers
Labelled report pairs	2,802 (651 positive, 11 pair types)
Trap groups	24 groups of confusable incidents
Evidence pairs (judge)	1,312 (1,156 real, 156 synthetic)

B. Hard Cases

Positive report pairs are cross-lingual (385), same-language (92) or delayed (174). Negative pairs range from easy to hard: random pairs (1,282), pairs on the same day (548), pairs at the same place with a different time or type (214), and adjacent houses on the same road that needed separate dispatches (49). The last kind, the *adjacent-house trap*, shares road, town, type and time and differs only in the house number. Four such pairs were placed in the test split so that this claim could be measured on unseen data.

C. Reference Labels

Reference fields and evidence-pair labels were drafted with the help of a large language model working from the caller's words alone, from a different model family than the runtime extractor, and were then checked in a blind human agreement study on 40 reports and 40 evidence pairs. Agreement was 89.4% on the fields and 77.5% on the evidence labels (Cohen's $\kappa = 0.57$), and all 12 decisive

conflicts were labelled the same. The check found systematic errors, corrected in new versions built by script: 208 people counts copied from the trapped count, and 699 evidence pairs labelled as agreeing only because the type matched.

D. Protocol

Every choice of model, hyper-parameter, threshold and feature set was made on the DEV split, and the rule that would decide each adoption was written before the run. TEST was scored once per adopted model. Every GPU-trained model had to pass a seven-check health report (a clean full-precision GPU run, decreasing loss, over-fitting a tiny subset, a shuffled-label control, beating the baseline, agreement of three seeds, no split overlap) and to be reproduced on the laptop CPU before use. Decisions are evaluated in an arrival-order replay: each report is compared only with what arrived before it, and the component builds its own incidents, so early mistakes carry forward. Fusion is scored by cross-fitting over five DEV folds, and the final DEV numbers are the mean over ten random fold divisions.

IV. System Architecture

Fig. 1 shows the nine processing steps. Step 1 turns a report into a reconciled record and Step 2 builds its evidence pool. Steps 3, 4, 5A and 5B are independent channels that compare the report with each candidate incident. Step 6 fuses them, Step 7 decides, Step 8 updates the incident and Step 9 flags ambiguity. The channels are trained separately and combined late.

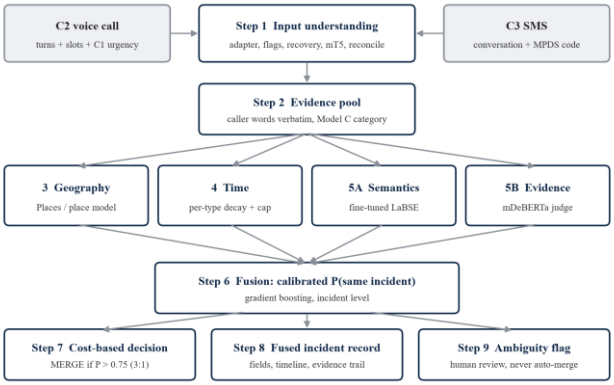


Fig. 1. The nine processing steps of the fusion component. Steps 3 to 5B are learned channels that are combined late by the fusion model.

A. Input Understanding and Evidence Pool

An adapter maps the call and SMS schemas into one record; a missing value is recorded as missing and never as zero, because "nobody is trapped" and "the caller did not say" must lead to different decisions. Fields are extracted by an mT5-small student [22] distilled from a guarded GPT-4o mini teacher (gpt-4o-mini-2024-07-18) that was used offline on the training split only. At run time no API is called; the student runs on the CPU in about three seconds, its location must be a span of the caller's own words, and any count must agree with its quote. The student, values recovered from the operator's questions and the upstream values are then reconciled field by field, with a confidence set by how many sources agree. Step 2 keeps one evidence item per caller turn, verbatim, paired with the operator question it answers; a logistic regression over LaBSE embeddings and character n-grams assigns a category. No translation is used, because a translation can remove the decisive house number, negation or place name.

B. Geography and Time

A Latin-script location is resolved with the Google Places API [24]. Native-script locations are kept as text, because a confident wrong coordinate is worse than none (Section VI-B). Two locations are then compared by a learned place model that outputs the probabilities of *same*, *near* and *far*. It uses frozen LaBSE embeddings and character n-gram TF-IDF of both strings with interaction features, reads Sinhala and Tamil through IAST transliteration so that cross-script pairs share characters, and is pre-trained with weight 0.3 on GeoNames [25] place pairs. A Platt-calibrated scorer combines distance, area and place-model outputs; a missing location gives a missing flag, never a zero score. For time, a logistic regression over a spline of the log gap is fitted per incident type: a report eight hours later belongs to the same flood with probability 0.086 but to the same fire with only 0.042. Geography and time rank the candidates, and the top K = 23 are passed on, K being the smallest cap that kept every true DEV link in a stress replay with all incidents open.

C. Semantic Similarity and Evidence Judge

Step 5A embeds the caller turns of each report with LaBSE [15] fine-tuned on the report pairs with a contrastive objective and hard negatives; operator turns are excluded because the operator asks the same questions in every call. Step 5B compares individual facts. Each evidence item is aligned with candidate items whose answers are similar (cosine ≥ 0.5, chosen on DEV), and a fine-tuned mDeBERTa-v3 [20] judge labels each aligned pair as *corroborates*, *contradicts*, *neutral* or *count_variance*, averaging both orders of the pair. The fourth label exists because witnesses often give different counts for one incident, which is not evidence of a different incident. The counts of each label and the maximum probabilities become features, and the aligned caller words are kept for the evidence trail.

D. Fusion and Decision

Step 6 builds one feature vector per report and candidate: the geography score and missing flag, place-model probabilities, the time score and log gap, type agreement, the semantic cosine, the judge summary and two house-number features. A histogram gradient boosting classifier of depth three outputs P(same). It is fitted on DEV, because the channel models were trained on TRAIN and their scores there are over-confident (semantic AUC 1.000 on TRAIN against 0.970 on DEV). Two kinds of added rows teach cases the data lack: synthetic adjacent-house reports, which are real reports with only the house number changed and processed through every step, and counterfactual [23] copies of same-incident rows with the type or house-number features flipped and labelled as different. Step 7 merges into the best candidate when

$$P(same) > C_{FM} / (C_{FM} + C_{FS}) = 0.75, \quad (1)$$

with a cost ratio $C_{FM} : C_{FS} = 3 : 1$ [7]; otherwise the report opens a new incident. A separated report whose best probability lies between 0.5 and the threshold is flagged for review. Two policy rules set by the researcher apply: a report without a location is never merged, and nothing is merged into an incident without a location. Step 8 makes the whole incident the unit of comparison: geography uses its best-matching member, time its latest member, and the judge its pooled evidence; the fused record keeps agreeing fields, count ranges, disagreeing values with their sources, a timeline and an evidence trail. Step 9 flags a report that scores above the threshold for two or more incidents. Urgency and priority are

shown to the dispatcher but are never inputs to the merge decision.

V. Experimental Setup

A. Implementation

The component is written in Python 3.11 with scikit-learn [26], PyTorch and Hugging Face Transformers [27]. The three transformer models were fine-tuned on a single NVIDIA T4 GPU on Google Colab, and all inference runs on a laptop CPU. External services are cached and switched off in every reported evaluation, so the results do not depend on a remote service. Every installed model has a provenance file with its training run, data hash and selection result.

B. Baselines and Metrics

Each channel is compared with the simplest alternative: character n-gram TF-IDF with logistic regression, an off-the-shelf encoder, or the earlier rule-based method. Channels are measured by ROC AUC on report pairs, the judge by macro-F1 over four labels, and fusion by log loss, Brier score, AUC and expected calibration error (ECE). The replay reports the false merge rate (share of all reports merged into a wrong incident), the false split rate (share of linkable reports, whose incident was already open, that were not merged into it), link recall, merge precision, the review rate, adjacent-trap merges and B-cubed precision and recall [28]. With 112 DEV and 115 TEST reports, one report moves a rate by about one percentage point.

VI. Results

A. Extraction and Evidence Pool

On the mean share of correct values over five fields, the mT5-small student scored 0.788 on DEV against 0.155 for the upstream extraction, with no field worse, and was adopted; on TEST it scored 0.73 against 0.12. An XLM-R span extractor reached only 0.50 on DEV. Table III shows the accuracy of the reconciled fields. The evidence category model reached a macro-F1 of 0.982 on DEV and 0.996 on TEST.

TABLE III. Reconciled Field Accuracy (%) Against Reference Fields

Split	Type	People	Trapped	Injured	Help
DEV	97.3	95.0	100	100	96.6
TEST	91.2	96.3	98.1	100	91.2

B. Geography (RQ1)

Google Places resolved 83% of Latin-script gold locations to within 1 km of the true place, but only 53% of Sinhala-script and 58% of Tamil-script locations, and placed 12–13% of native-script locations more than 25 km away, often confidently. The learned place model replaced a hand-written phonetic matcher (DEV macro-F1 0.777 against 0.720). Table IV gives the TEST AUC of every channel. Geography reached 0.973 when both reports had coordinates and 0.838 in text fallback, with pairs in which both locations are in native script remaining hardest at 0.787. Learned text comparison therefore supports linking where geocoding fails, although it is weaker than reliable coordinates.

TABLE IV. Channel Results on TEST

Channel	Measure	Baseline	Adopted
Geography (all pairs)	AUC	–	0.913
Geography (text fallback)	AUC	–	0.838
Time (arrival replay)	AUC	–	0.996
Semantic (5A)	AUC	0.846 / 0.621	0.946
Evidence judge (5B)	Macro-F1	–	0.718

For the semantic channel, the first baseline is off-the-shelf LaBSE and the second character TF-IDF.

C. Time and Semantic Similarity

The per-type time model was preferred to a single model on DEV (log loss 0.0535 against 0.0593). Ranking by geography and time placed the true link first for 94.7% of TEST queries and within the top three for all of them, and K = 23 kept 98.7% of true links in the TEST stress replay. The channels complement each other: two reports at the same place on different days scored 0.68 on geography but only 0.002 on time. The fine-tuned encoder passed all health checks (three seeds 0.972, 0.972, 0.970); removing near-translation pairs changed its TEST AUC only from 0.946 to 0.941, and by language it reached 0.972 for Singlish, 0.903 for Tanglish and 0.941 for native or mixed pairs.

D. Evidence Judge (RQ2)

The first judge run was rejected by its health report because its seeds disagreed (macro-F1 0.709, 0.820, 0.741). The second reached a TEST macro-F1 of 0.718 (F1 0.75 corroborates, 0.81 contradicts, 0.65 neutral, 0.67 count_variance). In live testing it read the house numbers of ‘23 Pansala Road’ and ‘14 Pansala Road’ as a count difference, so 156 synthetic pairs with only the house number changed were added. Table V shows the four-condition adoption rule written before the third run; all conditions were met.

TABLE V. Adoption Rule for the Third Judge Run (DEV)

Condition	Run 2	Run 3
Health and laptop checks pass	–	9/9
Macro-F1 (real rows) ≥ 0.7955	0.8055	0.8299
Synthetic rows improve	25/29	29/29
Report-pair AUC ≥ 0.9233	0.9283	0.9481

TEST, scored once after adoption, disagreed: macro-F1 fell from 0.718 to 0.687 and the report-pair AUC of the judge features from 0.915 to 0.886. The judge still adds information beyond semantic similarity (cross-validated DEV AUC 0.985 against 0.971; TEST 0.953 against 0.946 with the second run), and its contradictions fall where they matter: on 58% of adjacent-house trap pairs, against 2–5% of same-incident pairs and no easy negatives. The answer to RQ2 is therefore yes, with a smaller benefit on TEST than on DEV.

E. Fusion and Ablation

Gradient boosting fitted on DEV had the lowest cross-validated log loss (0.069 against 0.079 for logistic regression). On TEST the report-level fusion model reached a log loss of 0.175, a Brier score of 0.051, an AUC of 0.983 and an ECE of 0.033. Table VI removes one channel at a time and refits; every channel contributes, and the type features add nothing beyond the per-type time models.

TABLE VI. CHANNEL ABLATION (TEST LOG LOSS)

Configuration	Log loss
All channels	0.175
Without time	0.388
Without semantic similarity	0.232
Without evidence judge	0.202
Without geography	0.186

F. *Decision and Incident-Level Matching (RQ3)*

At report level with a 2:1 cost ratio, TEST gave a false merge rate of 0.9% but a false split rate of 24.0%; the synthetic negatives had taught the model that near-identical text with many corroborations is suspicious. Fixes to the fusion model all failed their DEV rules. Changing the unit of comparison to the whole incident passed, and was adopted with a 3:1 ratio. As Fig. 2 shows, on TEST the false split rate fell from 24.0% to 10.7% and link recall rose from 74.7% to 88.0%, with false merges at 1.7% (two reports, both adjacent-house traps) and B-cubed precision and recall of 0.980 and 0.897.

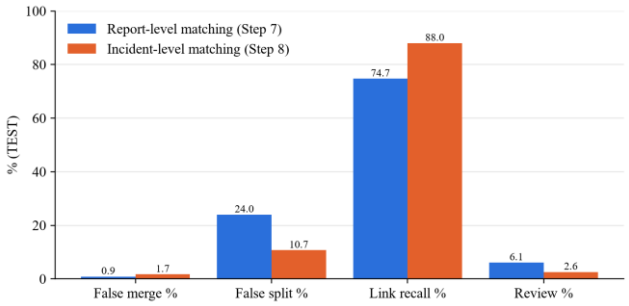


Fig. 2. Report-level against incident-level matching on TEST.

Two later changes came from live cases. Counterfactual type rows removed every fire–flood merge on DEV without a rule (flipped same-incident pairs above the threshold fell from 66 of 72 to 0 of 72), and house-number features lowered DEV false merges from 1.8% to 0.9%. For revision, merging two incidents retrospectively when a report scored above the threshold for both made four merges on DEV, three of them wrong, and was not adopted. Using the same signal to ask instead was adopted: on TEST, 6 of 115 reports were flagged and 4 of the 6 flags pointed at a real problem.

G. *Final Operating Point (RQ4)*

The selection had used one DEV fold division, which turned out to be the most favourable of ten, so Table VII gives DEV as the mean and standard deviation over ten divisions. TEST was scored once for this configuration.

TABLE VII. FINAL RESULTS OF THE LIVE CONFIGURATION

Measure (%)	DEV live	TEST live	TEST no rules
False merge	2.43 (0.70)	0.9	1.7
False split	19.73 (1.54)	28.0	14.7
Link recall	77.81 (1.84)	70.7	84.0
Merge precision	95.74 (1.24)	98.1	96.9
Review	8.55 (1.84)	6.1	6.1
Trap merges (count)	1.0 (0.78)	1	2
B-cubed F1	0.872	0.864	0.921

The live configuration keeps false merges at 0.9% of TEST reports with a merge precision of 98.1%, at the price of a high false split rate. Most of that price comes from the no-location rule: TEST has twelve reports without any location, and the rule keeps each of them, and anything that follows them, apart. Without the rules, link recall rises to 84.0% at 1.7% false merges. Raising the cost ratio lowers false merges slowly but raises false splits and review load quickly.

VII. IMPLEMENTATION AND INTEGRATION

The component runs as a Flask service. C2 and C3 publish each finished conversation through a versioned contract (c4.v2) validated with Pydantic; unknown fields are rejected with an explanatory error, so a change on either side cannot pass silently. The service runs Steps 1 to 3A and replies at once; a background worker then runs Steps 3B to 9 in arrival order and persists the incident store after every decision. This split was introduced after measurement: with the judge inside the reply, one report took about 12.7 s, close to the 20 s budget of the callers. The reply now takes about 4 s on the laptop CPU, and the judge finishes 0.9–2.3 s later in one batched call. With the extraction model switched off, the recovered and upstream values carried on and geography continued to work.

An operator page shows the output of every step for a report, and an incident dashboard shows key figures, a map, the channels that reached each incident, the decision confidence of every report and the fused record with its evidence trail. In a live session with all four components, 23 reports formed 17 incidents, five of which were reached by both calls and SMS. The automated suite of 421 tests checks invariants such as that a missing value stays missing, that evidence equals the caller's words character for character, that a count difference is never a contradiction and that the fusion features contain no urgency field. Each fault found live, such as an unread Sinhala teen number or a deictic phrase geocoded as a place, became a test and a fix measured on DEV.

VIII. DISCUSSION

The results support late fusion of independent channels because the channels fail on different cases. Geography cannot separate the same place on different days, which time separates; semantic similarity rises for reports with the same words (mean cosine 0.80 for same-place, different-day pairs), and the evidence judge finds the contradiction in adjacent-house traps. Incident-level matching was the most effective single design choice, more than halving false splits, which is consistent with the cluster-level comparison of event tracking [10]. A low false merge rate with lower automatic linking is an intentional trade-off: a merged emergency can be lost, whereas a split or flagged report remains visible to an operator.

The evaluation also showed how easily small-data results mislead. The judge improved on DEV but not on TEST, and the single DEV fold division used for selection was the most favourable of ten. Reporting the mean over ten divisions, keeping adoption rules fixed before each run and reporting unfavourable TEST results as found were therefore necessary, not optional.

A. *Limitations*

First, the corpus is simulated and written by the author; transfer to real hotline traffic is untested. Second, the splits are small, so differences of one or two points are within noise. Third, Tamil and Tanglish reference labels were not independently verified. Fourth, TEST had been scored for earlier configurations before the final one, although nothing was changed after the final run. Finally, transcripts are clean, so speech recognition errors are not modelled, and no study with dispatchers was carried out. Because no real calls were

used, no personal data were processed; API keys are kept only in an environment file, and the caller's words are never rewritten by a model.

IX. Conclusion

This paper presented a component that decides whether multilingual emergency calls and SMS reports describe the same incident. It keeps the caller's words verbatim, scores each report against the open incidents on four learned channels, fuses them into a calibrated probability and merges only above a cost-derived threshold. On a purpose-built corpus with development-only selection, the live configuration wrongly merged 0.9% of test reports with a merge precision of 98.1% and linked 70.7% of repeat reports, or 84.0% at 1.7% false merges without the no-location rule. Learned cross-script place comparison worked where geocoding of native script did not, the evidence judge separated adjacent houses that other channels confused, and incident-level matching cut false splits from 24.0% to 10.7%. The most important next step is validation on real, anonymised hotline reports held out as an external test set, together with a study of how dispatchers use the review flags.

References

[1] International Academies of Emergency Dispatch, "Medical Priority Dispatch System (MPDS)," Salt Lake City, UT, USA. [Online]. Available: https://www.emergencydispatch.org

[2] I. P. Fellegi and A. B. Sunter, "A theory for record linkage," J. Amer. Statist. Assoc., vol. 64, no. 328, pp. 1183–1210, 1969.

[3] A. K. Elmagarmid, P. G. Ipeirotis, and V. S. Verykios, "Duplicate record detection: A survey," IEEE Trans. Knowl. Data Eng., vol. 19, no. 1, pp. 1–16, 2007.

[4] P. Christen, Data Matching: Concepts and Techniques for Record Linkage, Entity Resolution, and Duplicate Detection. Berlin, Germany: Springer, 2012.

[5] J. C. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in Advances in Large Margin Classifiers. Cambridge, MA, USA: MIT Press, 1999, pp. 61–74.

[6] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in Proc. 34th Int. Conf. Mach. Learn. (ICML), 2017, pp. 1321–1330.

[7] C. Elkan, "The foundations of cost-sensitive learning," in Proc. 17th Int. Joint Conf. Artif. Intell. (IJCAI), 2001, pp. 973–978.

[8] A. Cybulska and P. Vossen, "Using a sledgehammer to crack a nut? Lexical diversity and event coreference resolution," in Proc. 9th Int. Conf. Lang. Resour. Eval. (LREC), 2014, pp. 4545–4552.

[9] K. Lee, L. He, M. Lewis, and L. Zettlemoyer, "End-to-end neural coreference resolution," in Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP), 2017, pp. 188–197.

[10] J. Allan, J. Carbonell, G. Doddington, J. Yamron, and Y. Yang, "Topic detection and tracking pilot study: Final report," in Proc. DARPA Broadcast News Transcription and Understanding Workshop, 1998, pp. 194–218.

[11] S. Vieweg, A. L. Hughes, K. Starbird, and L. Palen, "Microblogging during two natural hazards events: What Twitter may contribute to situational awareness," in Proc. SIGCHI Conf. Human Factors Comput. Syst. (CHI), 2010, pp. 1079–1088.

[12] M. Imran, C. Castillo, F. Diaz, and S. Vieweg, "Processing social media messages in mass emergency: A survey," ACM Comput. Surv., vol. 47, no. 4, pp. 1–38, 2015.

[13] A. Conneau et al., "Unsupervised cross-lingual representation learning at scale," in Proc. 58th Annu. Meeting Assoc. Comput. Linguistics (ACL), 2020, pp. 8440–8451.

[14] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. EMNLP-IJCNLP, 2019, pp. 3982–3992.

[15] F. Feng, Y. Yang, D. Cer, N. Arivazhagan, and W. Wang, "Language-agnostic BERT sentence embedding," in Proc. 60th Annu. Meeting Assoc. Comput. Linguistics (ACL), 2022, pp. 878–891.

[16] N. de Silva, "Survey on publicly available Sinhala natural language processing tools and research," arXiv:1906.02358, 2019.

[17] B. R. Chakravarthi, V. Muralidaran, R. Priyadharshini, and J. P. McCrae, "Corpus creation for sentiment analysis in code-mixed Tamil-English text," in Proc. SLTU-CCURL Workshop, 2020, pp. 202–210.

[18] S. R. Bowman, G. Angeli, C. Potts, and C. D. Manning, "A large annotated corpus for learning natural language inference," in Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP), 2015, pp. 632–642.

[19] A. Conneau et al., "XNLI: Evaluating cross-lingual sentence representations," in Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP), 2018, pp. 2475–2485.

[20] P. He, J. Gao, and W. Chen, "DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing," in Proc. 11th Int. Conf. Learn. Represent. (ICLR), 2023.

[21] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," arXiv:1503.02531, 2015.

[22] L. Xue et al., "mT5: A massively multilingual pre-trained text-to-text transformer," in Proc. NAACL-HLT, 2021, pp. 483–498.

[23] D. Kaushik, E. Hovy, and Z. C. Lipton, "Learning the difference that makes a difference with counterfactually-augmented data," in Proc. 8th Int. Conf. Learn. Represent. (ICLR), 2020.

[24] Google, "Places API documentation," Google Maps Platform. [Online]. Available: https://developers.google.com/maps/documentation/places/web-service

[25] GeoNames, "GeoNames geographical database," CC-BY 4.0. [Online]. Available: https://www.geonames.org

[26] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.

[27] T. Wolf et al., "Transformers: State-of-the-art natural language processing," in Proc. EMNLP: System Demonstrations, 2020, pp. 38–45.

[28] A. Bagga and B. Baldwin, "Algorithms for scoring coreference chains," in Proc. LREC Workshop on Linguistic Coreference, 1998, pp. 563–566.